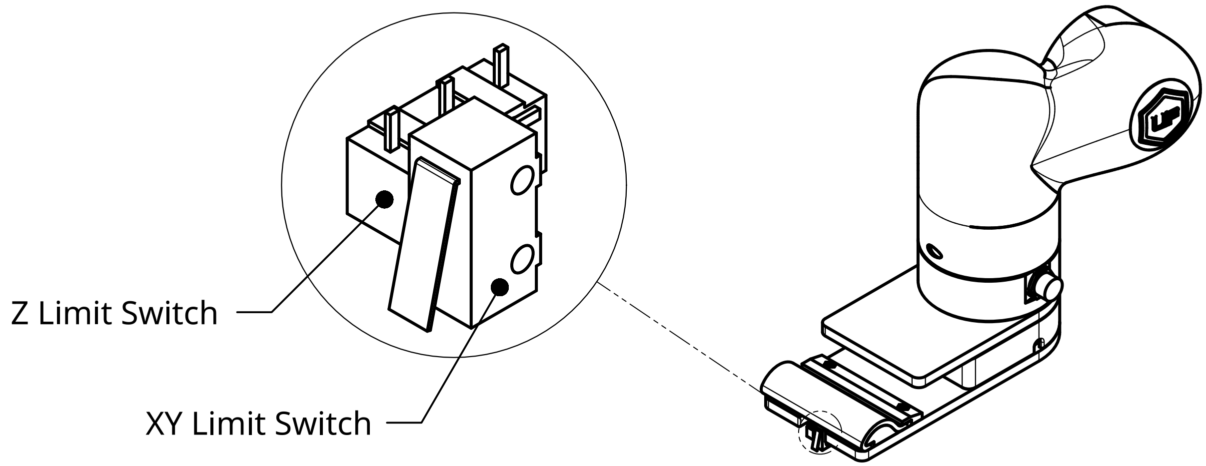
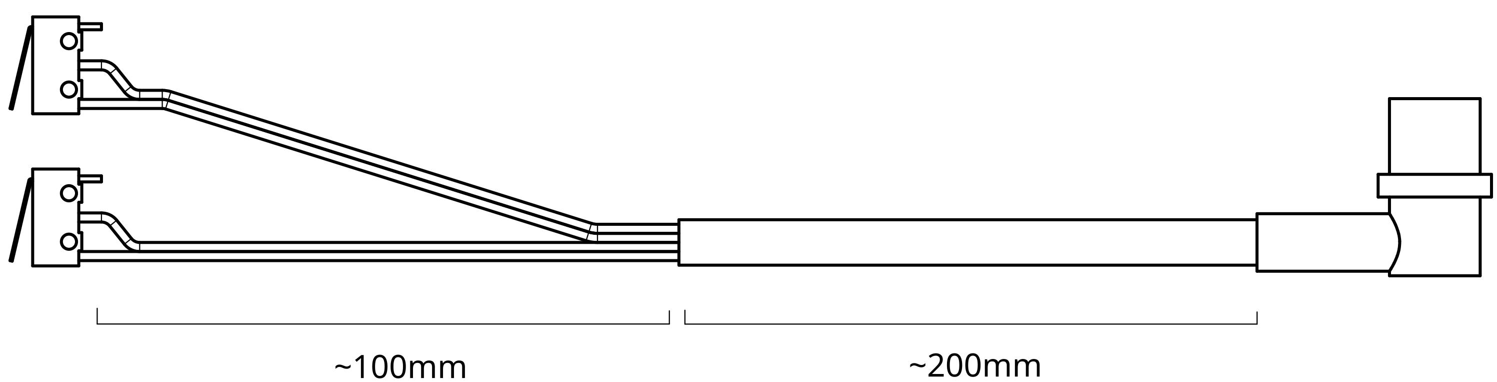
Limit Switch Wiring

The end effector contains two limit switches that should be wired prior to assembly. Both of the switches are wired normally open, with approximate wire lengths shown below.

Referencing the xArm wiring diagram (provided on page 2), the Z limit switch should be wired to power and TI0, and the XY limit switch should be wired to power and TI1.

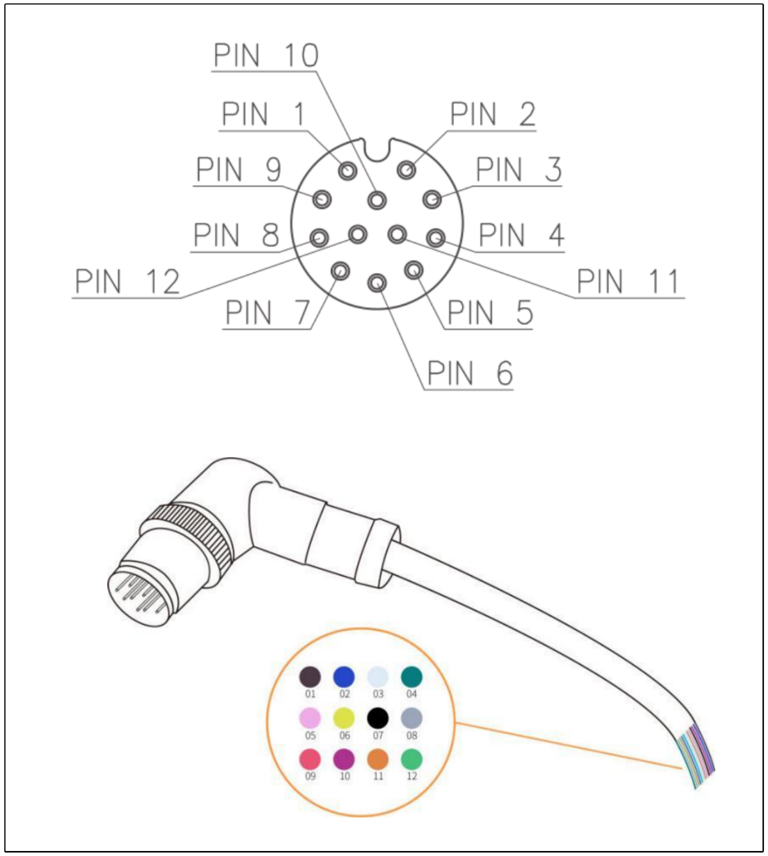


Note: wiring is based off of the 12 pin cable provided with the xArm6; there is a thicker section of bundled wires ~200mm in length split off into smaller guage wires ~100mm in length.



Wiring Reference

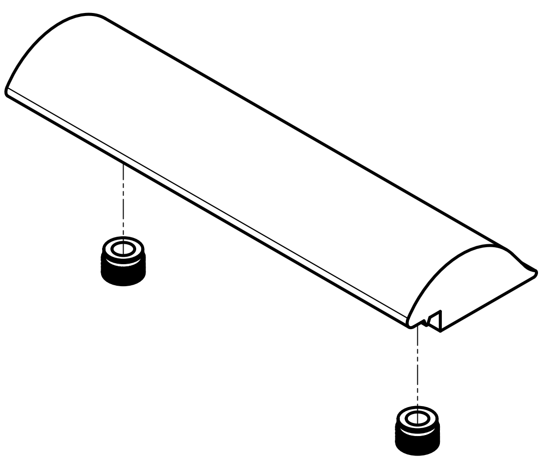
Pin	Color	Signal
1	Brown	+24V (Power)
2	Blue	+24V (Power)
3	White	0V (GND)
4	Green	0V (GND)
5	Pink	User 485-A
6	Yellow	User 485-B
7	Black	Tool Output 0 (TO0)
8	Grey	Tool Output 1 (TO1)
9	Red	Tool Input 0 (TI0)
10	Purple	Tool Input 1 (TI1)
11	Orange	Analog Input 0 (AI0)
12	Light Green	Analog Input 1 (AI1)

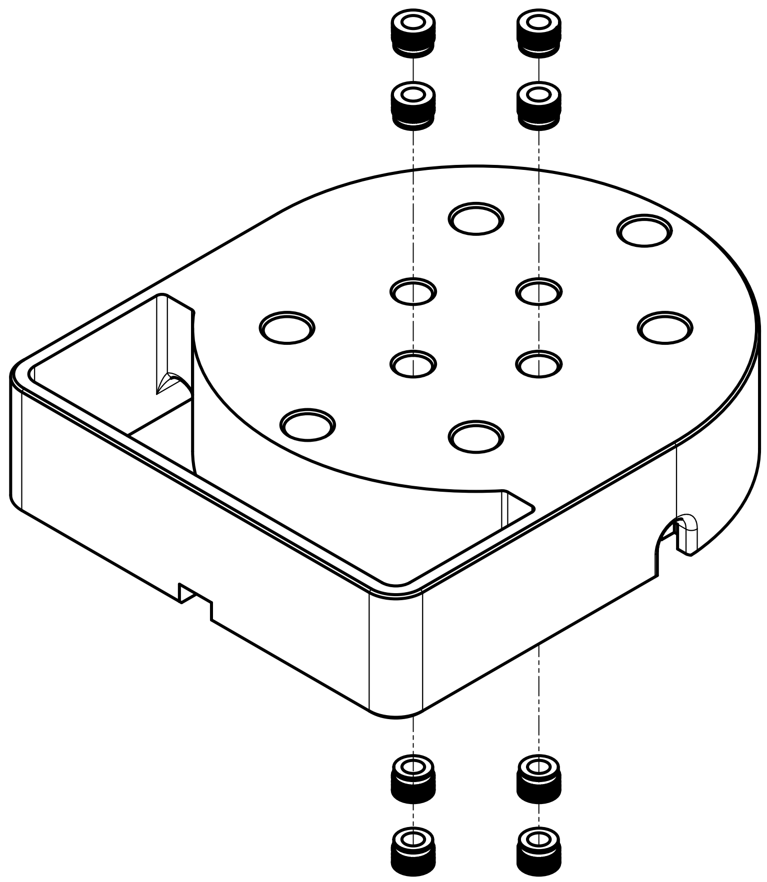


Wiring info from xArm User Manual.

Heatset Inserts

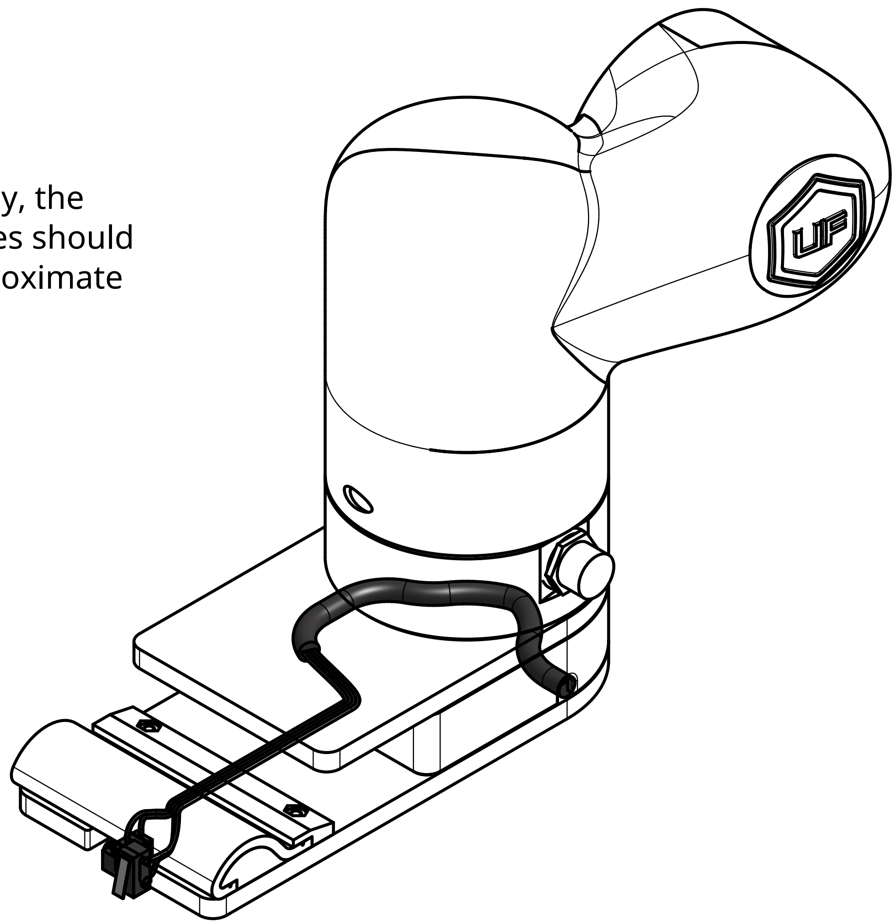
The end effector contains six heat set inserts: two in the front clamp and four in the middle insert.

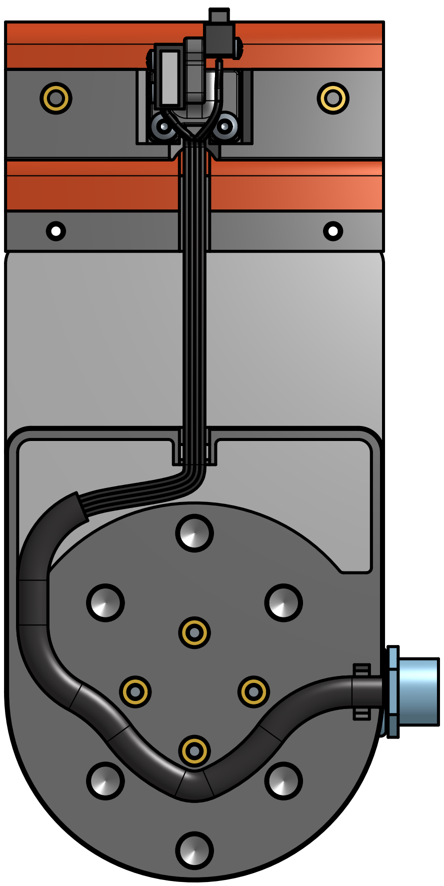




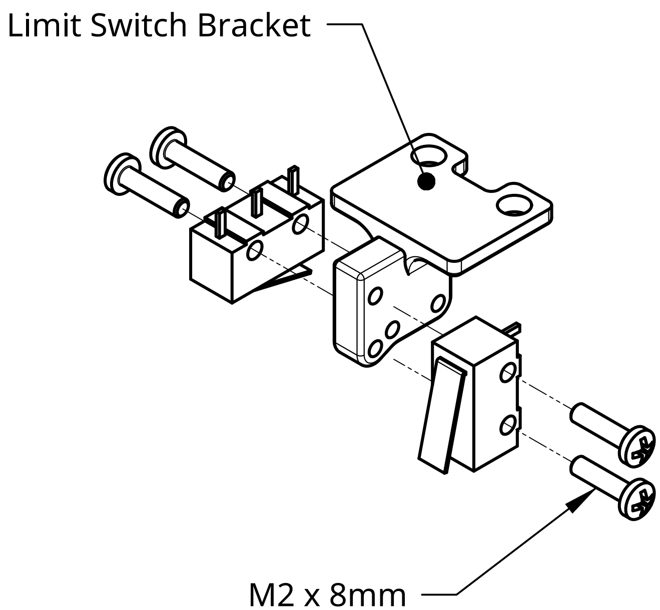
Wiring Layout

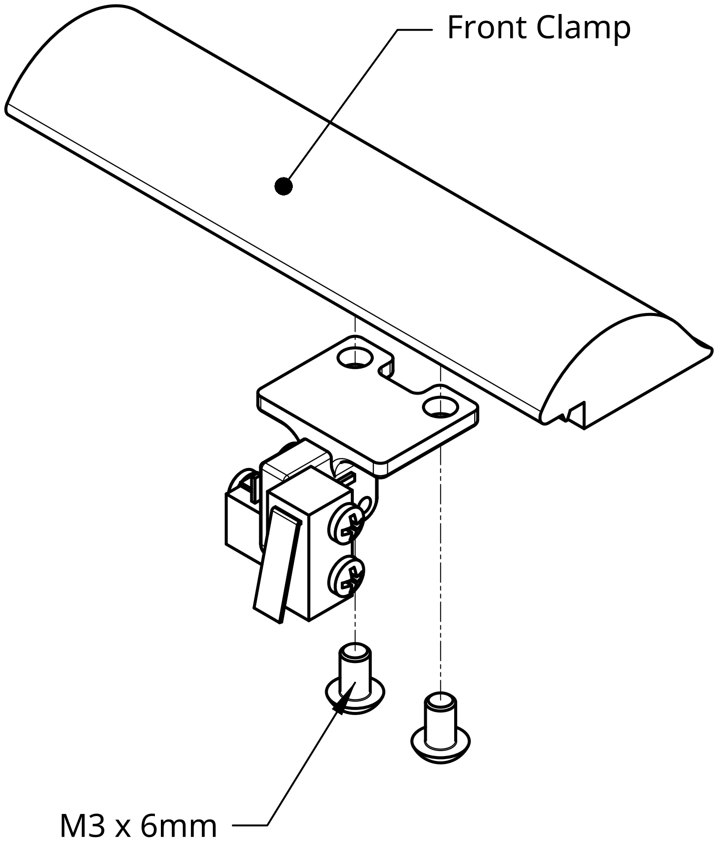
During assembly, the limit switch wires should follow this approximate path.



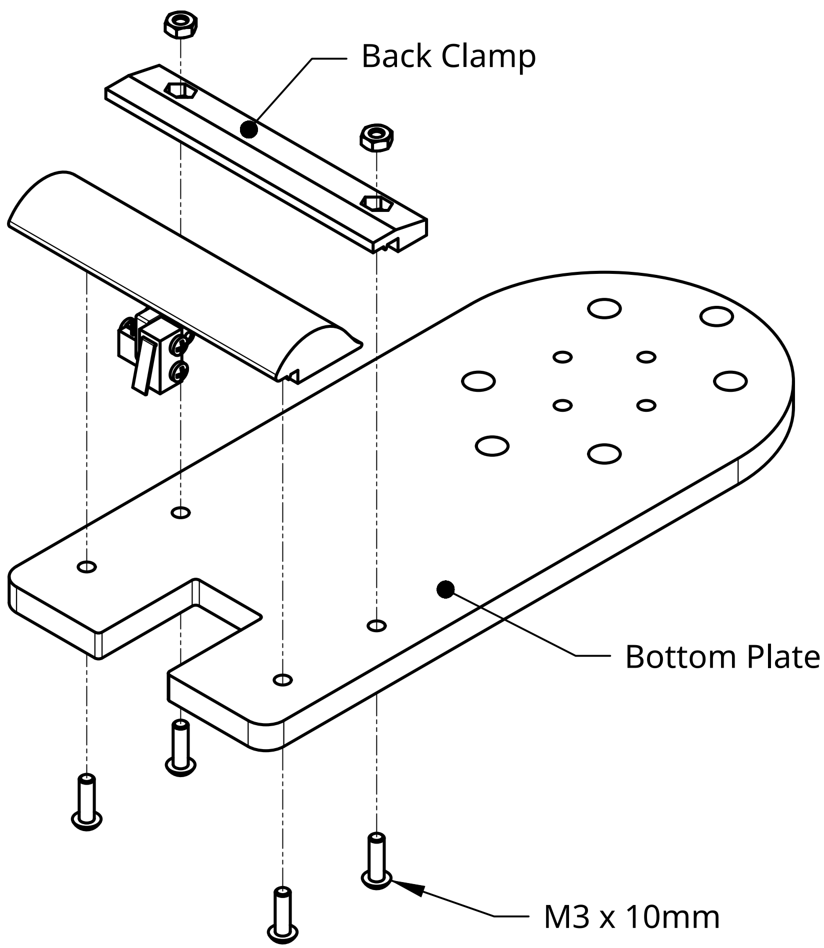


Assembly

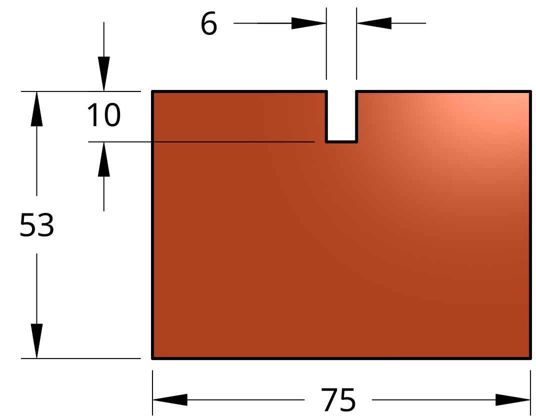


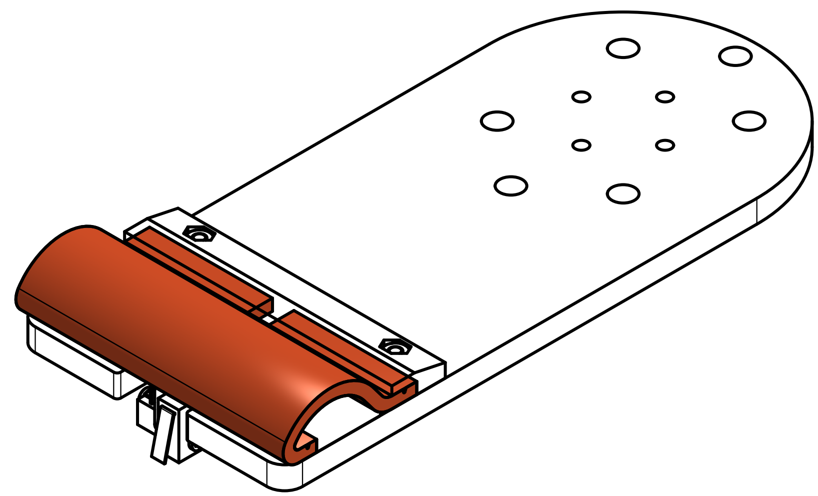


Assembly

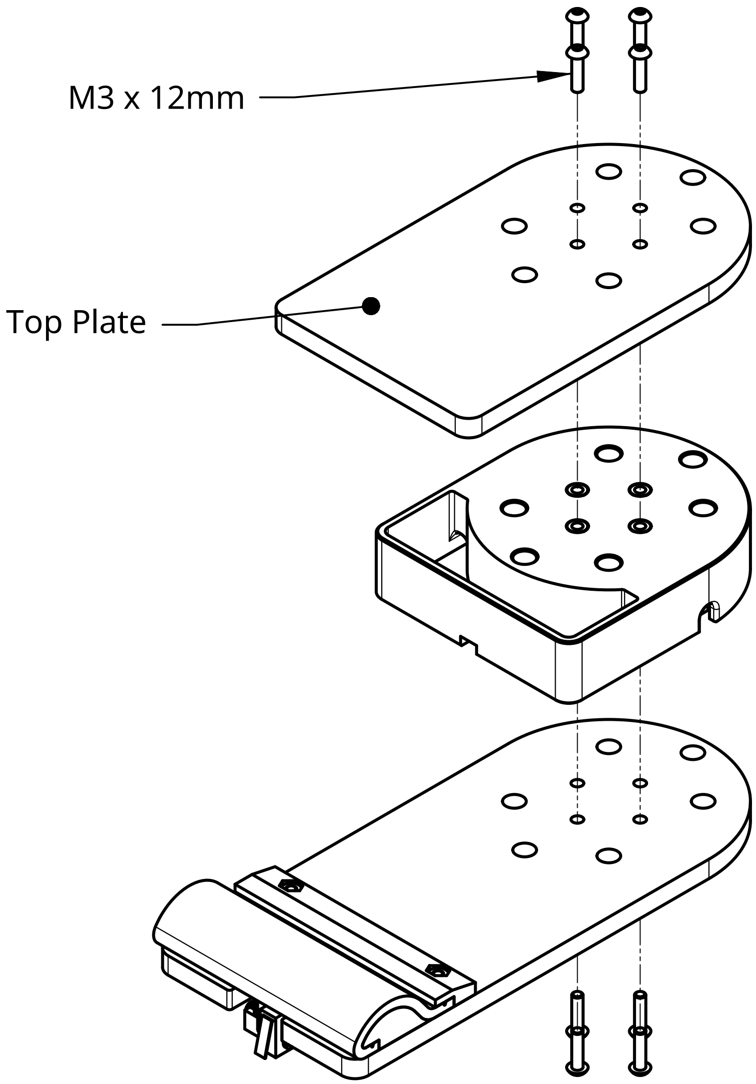


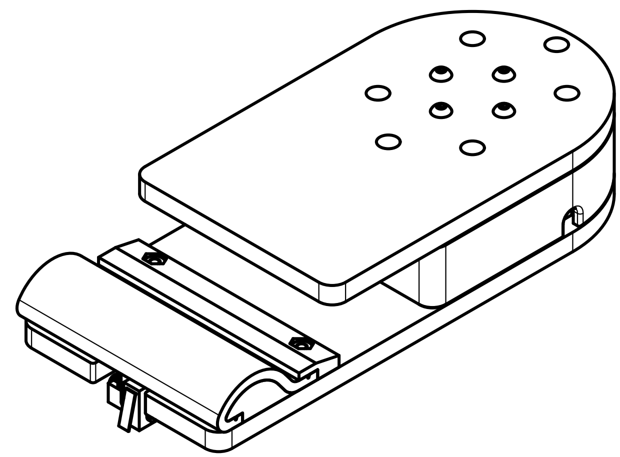
A ~3 mm thick silicon pad with dimensions shown to the right should be clamped using the 3D printed parts.





Assembly





Assembly